

Leflunomide Treatment in Elderly Patients with Rheumatoid or Psoriatic Arthritis

Retrospective Analysis of Safety and Adherence to Treatment

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Abstract

Background: Disease-modifying antirheumatic drugs (DMARDs) play a crucial role in the treatment of persistent chronic synovitis, such as active rheumatoid arthritis (RA) and spondyloarthritis, by inducing or maintaining disease remission, reducing the frequency of flares or relapses, and allowing corticosteroids to be tapered while maintaining disease control.

Objective: The aim of this retrospective study was to evaluate the safety of, and adherence to treatment with, leflunomide in elderly RA and psoriatic arthritis patients compared with younger patients.

Methods: A total of 90 Italian patients (80 with active RA and 10 with psoriatic arthritis) were retrospectively examined at entry and after 24 months' follow-up. Patients were divided into two groups according to age: those aged ≤ 65 years ($n = 50$) and those aged > 65 years ($n = 40$). Each patient was analysed for clinical, demographic and laboratory parameters in order to evaluate liver, renal and haematological toxicity. Disease Activity Score including a 28-joint count (DAS28) and physician global assessment of disease activity (MD global) were measured to define disease activity.

Results: During the 24-month follow-up period, 30 patients (33.3%) discontinued leflunomide: 17 patients (34.0%) in the group of patients aged ≤ 65 years and 13 patients (32.5%) in those aged > 65 years. There were no differences in treatment withdrawal between the two groups. Overall, 10 patients (11.1%) in the entire study population discontinued leflunomide for lack of efficacy, while 21 (23.3%) discontinued the drug because of adverse effects (one patient withdrew because of both inefficacy and adverse effects). There were no significant differences in efficacy or adverse effects between patients aged ≤ 65 years and patients aged > 65 years. There was also no significant difference in survival rates of leflunomide treatment when patients aged ≤ 65 years were compared with patients aged > 65 years ($p = 0.94$).

There were no significant differences in withdrawal rates in the overall population when leflunomide monotherapy was compared with leflunomide combination therapy. There were also no significant differences in the types

of adverse effects associated with monotherapy or combination therapy when the two age groups were compared.

Conclusions: Leflunomide is a useful and well tolerated DMARD for the treatment of RA and psoriatic arthritis in the elderly. The safety profile of, and adherence to, leflunomide is not different in older patients with chronic inflammatory joint diseases such as RA or psoriatic arthritis to that observed in younger patients.

Background

In elderly patients with arthritis, pain, stiffness and systemic symptoms contribute to weakness, immobility and increased frequency of falls, which may decrease independence and reduce lifespan. Treatment with disease-modifying antirheumatic drugs (DMARDs) should be started within 3 months of diagnosis in patients with new onset persistent chronic synovitis, such as active rheumatoid arthritis (RA) and spondyloarthritis, because of associated early irreversible joint damage.^[1] Immunosuppressive drugs are used in the treatment of RA and several rheumatological conditions to induce or maintain disease remission, reduce the frequency of flares or relapses, and permit tapering of corticosteroids while maintaining disease control. Leflunomide is an immunosuppressive drug that reduces the proliferation and activity of lymphocytes by inhibiting dihydroorotate dehydrogenase, a key enzyme in *de novo* synthesis of pyrimidine-containing ribonucleotides.^[2] Hypertension, gastrointestinal symptoms (such as nausea and diarrhoea), elevations of serum aminotransferases and leukopenia are its major potential adverse effects.^[3-5] In this study we retrospectively evaluated the adherence to, and safety of, treatment with leflunomide in elderly RA and psoriatic arthritis patients compared with younger patients.

Methods

We retrospectively analysed 90 patients attending the Rheumatology Clinic of the Catholic University of Sacred Heart in Rome between January 2005 and July 2007. A total of 80 (88.9%) patients had active RA and met the 1987 Amer-

ican College of Rheumatology (ACR) criteria for the classification of RA, and 10 (11.1%) patients had active psoriatic arthritis. Our cohort was divided into two different groups according to age: 50 patients ≤ 65 years (group A) and 40 patients > 65 years (group B).

All patients were treated with leflunomide 20 mg/day, were taking NSAIDs and/or low doses of prednisone (≤ 5 mg/day) and began leflunomide without a loading dose.

The time of leflunomide first administration was considered the inclusion time in the analysis (T0), after which each subject was followed up clinically for at least 24 months. The following data were recorded for each patient: demographic characteristics, medical records, age at onset of RA or psoriatic arthritis, time of administration, duration of leflunomide therapy, reasons for its withdrawal, and concomitant DMARDs or tumour necrosis factor (TNF)- α inhibitor therapy. Each patient underwent, during the follow-up period, blood tests to assess liver function (transaminases, γ -glutamyl transpeptidase, alkaline phosphatase, lactate dehydrogenase [LDH]), renal function (serum creatinine and urine tests) and haematological toxicity (complete blood count). Toxicity was defined as a ≥ 3 -fold increase in transaminase levels (for hepatic toxicity) and a 50% increase in serum creatinine compared with baseline or urinary sedimentation abnormality (for renal toxicity). Haematological toxicity was defined as leukocytes $< 3.0 \times 10^9/L$, neutrophils ≥ 1.5 to $< 2.0 \times 10^9/L$, lymphocytes $< 1.0 \times 10^9/L$, haemoglobin < 10.0 g/dL or platelets $< 75.0 \times 10^9/L$. During the follow-up period, disease activity was measured by the Disease Activity Score including a 28-joint count (DAS28)^[6] and on the physician global assessment of disease activity

(MD global)^[7] recorded on a 10 cm visual analogue scale. In the analysis, patients with persistent disease activity, as shown by a DAS28 ≥3.2 and MD global score >1.5 cm, stopped treatment because of lack of efficacy.

The primary endpoints of the study were to: (i) assess the putative differential role of age in the occurrence of adverse effects leading to withdrawal of, or poor response to, leflunomide; (ii) assess (by analysing discontinuations and reported adverse effects) the tolerability of leflunomide; and (iii) assess the effect on tolerability of concomitant medications such as DMARDs and biological therapies.

Data were analysed using SPSS 12.0 (SPSS Inc., Chicago, IL, USA) for Windows and Prism® Software (GraphPad Software, San Diego, CA, USA). Categorical and quantitative variables were respectively described as frequencies, percentages and means ± standard deviation (SD). The Mann-Whitney test was used to compare variables between two groups. Categorical variables were analysed using chi-squared (χ^2) test or Fisher's Exact test, and odds ratios (ORs) with 95% confidence intervals (CIs) were calculated. Kaplan-Meyer survival analysis and the log-rank test were used to investigate the relevance of age (>65 years) to leflunomide discontinuation during 24-month follow-up. A value of $p < 0.05$ was considered statistically significant.

Results

Ninety patients (80 with a diagnosis of RA and 10 with psoriatic arthritis) were included in the analysis (mean ± SD age 59.69 ± 14.5 years and mean ± SD disease duration 11.41 ± 10.0 years), of whom 71 (79%) were women. The main demographic features and the discontinuation rates for leflunomide in the cohort of patients are shown in table I. The mean ± SD duration of leflunomide therapy was 18.60 ± 24.0 months for the entire cohort. During the 24-month follow-up period, 30 patients (33.3%) discontinued leflunomide: 10 (11.1%) discontinued leflunomide because of lack of efficacy, while 21 (23.3%) experienced adverse effects leading to drug discontinuation (one patient withdrew because of both ineffi-

Table I. Demographic characteristics and leflunomide discontinuation rates for the overall study population (n = 90)

Parameter	Value
Female [n (%)]	71 (79)
Age [y (mean ± SD)]	59.69 ± 14.5
Diagnosis [n (%)]	
rheumatoid arthritis	80 (89)
psoriatic arthritis	10 (11)
Immunological markers [n (%)]	
anti-CCP antibody positivity	30/52 (57.6)
IgA rheumatoid factor positivity	23/48 (47.9)
IgM rheumatoid factor positivity	27/48 (56.3)
Disease duration [y (mean ± SD)]	11.41 ± 10.0
Therapy duration [mo (mean ± SD)]	18.60 ± 24.0
Total drug discontinuations [n (%)]	30 (33.3)
lack of efficacy	10 (11.1)
adverse effects	21 (23.3)
arterial hypertension ^a	4 (4.4)
gastrointestinal ^b	10 (11.1)
cutaneous ^c	8 (8.8)

a New onset.
b Diarrhoea, hepatotoxicity, nausea, gastric pain, vomiting.
c Rash, hair loss, angioedema.
CCP = cyclic citrullinated peptide.

cacy and adverse effects). Nineteen (21.1%) of all patients had a history of arterial hypertension and 4 (4.4%) patients developed this condition during leflunomide treatment. Ten (11.1%) patients experienced gastrointestinal adverse effects and 8 (8.8%) had cutaneous manifestations that led to treatment interruption.

Table II summarizes the main demographic characteristics of the two groups of patients divided by age. There were no differences in terms of disease activity, as measured by the DAS28, when the two groups were compared at the beginning of therapy and after 24 months' follow-up. There were also no significant differences between the two groups in terms of the other main demographic features or in terms of both disease and treatment durations. There was no significant difference in the drug withdrawal rate between the two groups: 17 patients (34.0%) in group A and 13 patients (32.5%) in group B stopped leflunomide treatment. Skin rash and hair loss occurred in 8.8% of the overall population

without any significant difference between younger and older patients (8.0% and 10.0%, respectively).

When only patients with the diagnosis of RA were considered, again there were no differences in terms of disease activity, as measured by the DAS28, when the two groups of older and younger patients were compared at the beginning of therapy and after 24 months' follow-up (table III). Moreover, there were no significant differences in types of adverse effects between the two groups.

Analysis of the reasons for leflunomide treatment interruption showed that skin rash, hepatotoxicity and gastrointestinal adverse effects were the most frequent causes within the first 3 months of treatment (4.4%, 3.8% and 3.8%, respectively), while lack of efficacy and gastrointestinal adverse effects were the major causes after 10 months of treatment.

Univariate analysis showed that age, sex and IgA and IgM rheumatoid factor positivities were associated with p -values < 0.25 , and these factors were consequently included in a multivariate analysis. None of these variables achieved statistical significance as a factor linked to leflunomide withdrawal after 24 months' follow-up in the entire cohort of patients.

Analysis of withdrawal rate by therapeutic regimen showed that, for the overall study population, 20 (38.5%) patients of the 52 receiving monotherapy with leflunomide discontinued the drug during 24 months of follow-up. This compared with 10 (26.3%) of 38 patients who took leflunomide in combination with other drugs such as methotrexate (18.9%), TNF α inhibitors (13.3%) and DMARDs other than methotrexate (10.0%). There were no significant differences in leflunomide withdrawal rates between patients

Table II. Demographic characteristics and leflunomide discontinuation rates in younger versus older patient groups

Parameter	Group A (age ≤ 65 y) [n = 50]	Group B (age > 65 y) [n = 40]	p-Value
Female [n (%)]	38 (76)	33 (82.5)	NS
Age [y (mean $\pm$ SD)]	48.6 $\pm$ 10.2	73.5 $\pm$ 4.7	
Diagnosis [n (%)]			
rheumatoid arthritis	41 (82.0)	39 (97.5)	
psoriatic arthritis	9 (18.0)	1 (2.5)	
Disease duration [y (mean $\pm$ SD)]	11.2 $\pm$ 7.6	11.7 $\pm$ 9.0	NS
DAS28 score (mean $\pm$ SD) ^a			
baseline	3.6 $\pm$ 0.8	3.5 $\pm$ 0.7	NS
at last evaluation	2.9 $\pm$ 1.0	3.0 $\pm$ 0.9	NS
Immunological markers [n (%)]			
anti-CCP antibody positivity	16/29 (55.2)	14/23 (60.9)	NS
IgA rheumatoid factor positivity	13/29 (44.8)	10/19 (52.6)	NS
IgM rheumatoid factor positivity	15/29 (51.7)	12/19 (63.2)	NS
Therapy duration [mo (mean $\pm$ SD)]	18.45 $\pm$ 8.9	18.78 $\pm$ 9.1	NS
Total drug discontinuations [n (%)]	17 (34.0)	13 (32.5)	NS
lack of efficacy	6 (12.0)	4 (10.0)	NS
adverse effects	12 (24.0)	9 (22.5)	NS
arterial hypertension ^b	2 (4.0)	2 (5.0)	NS
gastrointestinal ^c	6 (12.0)	4 (10.0)	NS
cutaneous ^d	4 (8.0)	4 (10.0)	NS

a DAS28 score evaluated in 63 patients (32 in group A and 31 in group B).

b New onset.

c Diarrhoea, hepatotoxicity, nausea, gastric pain, vomiting.

d Rash, hair loss, angioedema.

CCP = cyclic citrullinated peptide; **DAS28** = Disease Activity Score including a 28-joint count; **NS** = not significant.

Table III. Demographic characteristics and leflunomide discontinuation rates in younger versus older rheumatoid arthritis patients

Parameter	Group A (age ≤65 y) [n = 41]	Group B (age >65 y) [n = 39]	p-Value
Female [n (%)]	32 (78)	33 (84.6)	NS
Age [y (mean ± SD)]	48.7 ± 11.1	73.6 ± 4.8	
Disease duration [y (mean ± SD)]	12.7 ± 8.2	12.5 ± 9.6	NS
Immunological markers [n (%)]			
anti-CCP antibody positivity	15/24 (62.5)	14/23 (60.9)	NS
IgA rheumatoid factor positivity	13/24 (54.2)	10/19 (52.6)	NS
IgM rheumatoid factor positivity	15/24 (62.5)	12/19 (63.2)	NS
DAS28 score (mean ± SD) ^a			
baseline	3.5 ± 0.7	3.4 ± 0.5	NS
at last evaluation	2.9 ± 0.6	3.0 ± 0.6	NS
Total drug discontinuations [n (%)]	11 (26.8)	13 (33.3)	NS
lack of efficacy	3 (7.3)	4 (10.3)	NS
adverse effects	11 (26.8)	9 (23.1)	NS

a DAS28 score evaluated in 53 patients (23 in group A and 30 in group B) at follow-up.

CCP = cyclic citrullinated peptide; **DAS28** = Disease Activity Score including a 28-joint count; **NS** = not significant.

taking leflunomide monotherapy and those taking leflunomide in combination with other therapy in the overall population ($p=0.22$; table IV).

In group A, 10 (38.5%) of the 26 patients receiving monotherapy with leflunomide and 7 (29.2%) of the 24 patients taking combination therapy discontinued treatment, without any significant difference between the two groups ($p=0.56$). In group B, 10 (38.5%) of the 26 patients receiving monotherapy with leflunomide and 3 (21.4%) of the 14 patients taking combination therapy discontinued treatment, without any significant differences between the two groups ($p=0.32$; table V).

The results of an analysis of the reasons for leflunomide treatment discontinuation in patients taking different therapeutic strategies are shown in table VI.

A similar survival rate on treatment was confirmed by the log-rank test when group A and group B were compared ($p=0.94$).

Discussion

There has been increasing interest in evaluating the prognostic and therapeutic implications of RA in elderly patients because the prevalence and impact of rheumatic conditions are predicted to increase in future years as a consequence of the

increased longevity of the general population. People >65 years of age comprise the fastest growing subpopulation and constitute an increasing proportion of the general population compared with other age groups; furthermore, approximately one-third of all RA patients are >65 years of age.^[8]

Studies conducted in Sweden and the US have shown that increasing age is an important

Table IV. Leflunomide discontinuation in patients taking leflunomide as monotherapy or in combination with other therapies in the overall study population

Treatment	Study population [n (%)] (n = 90)
Leflunomide monotherapy	52 (57.8)
Drug discontinuations	20 (38.5)
Combination therapy	38 (42.2)
Drug discontinuations	10 (26.3)
<i>Leflunomide and methotrexate</i>	<i>17 (18.9)</i>
Drug discontinuations	3 (17.6)
<i>Leflunomide and other DMARDs^a</i>	<i>9 (10)</i>
Drug discontinuations	2 (22.2)
<i>Leflunomide and TNFα inhibitors</i>	<i>12 (13.3)</i>
Drug discontinuations	5 (41.7)

a DMARDs other than methotrexate (i.e. chloroquine, ciclosporin and salazopyrine).

DMARDs = disease-modifying antirheumatic drugs; **TNF** = tumour necrosis factor.

Table V. Leflunomide discontinuation in patients taking leflunomide as monotherapy or in combination with other therapies in younger versus older patient groups

Treatment	Group A (age ≤65 y) [n (%)] (n = 50)	Group B (age >65 y) [n (%)] (n = 40)	p-Value
Leflunomide monotherapy	26 (52.0)	26 (65.0)	NS
Drug discontinuations	10 (38.5)	10 (38.5)	NS
Combination therapy	24 (48.0)	14 (35.0)	NS
Drug discontinuations	7 (29.2)	3 (21.4)	NS
<i>Leflunomide and methotrexate</i>	<i>12 (24.0)</i>	<i>5 (12.0)</i>	<i>NS</i>
Drug discontinuations	2 (16.7)	1 (20)	NS
<i>Leflunomide and other DMARDs^a</i>	<i>3 (6.0)</i>	<i>6 (15)</i>	<i>0.05</i>
Drug discontinuations	1 (33.3)	1 (16.7)	NS
<i>Leflunomide and TNFα inhibitors</i>	<i>9 (18.0)</i>	<i>3 (7.5)</i>	<i>NS</i>
Drug discontinuations	4 (44.4)	1 (33.3)	NS

a DMARDs other than methotrexate (i.e. chloroquine, ciclosporin and salazopyrine).

DMARDs = disease-modifying antirheumatic drugs; **NS** = not significant; **TNF** = tumour necrosis factor.

determinant of less aggressive RA care.^[9,10] Compared with patients aged 65–74 years, patients aged 75–84 years were 52% less likely to receive DMARDs and patients aged >85 years were 74% less likely.^[9]

Leflunomide, one of the most recently approved DMARDs, has been shown in multicentre, comparative studies to be as effective as the two most commonly used DMARDs – salazopyrine and methotrexate.^[11,12] Leflunomide has shown significant efficacy in the treatment of RA, psoriatic arthritis and ankylosing spondylitis.^[11–14]

Cohen et al.^[11] reported 85% survival on treatment for leflunomide and 79% for methotrexate in non-elderly patients with RA after 24 months’ follow-up. Thus, compared with methotrexate survival on treatment as reported in the literature, leflunomide shows a similar capacity to maintain disease control without evidence of new or increased toxicity over a 24-month follow-up. In our study, survival on treatment after 24 months’ follow-up was much lower (66.3%) than that reported in the Cohen et al.^[11] trial, thus confirming that ‘real world’ situations are not as favourable as the ideal world of the clinical trial; however, no difference was seen between elderly and younger arthritis patients.

In our retrospective follow-up study, the most common primary adverse events were gastrointestinal symptoms (including hepatotoxicity), skin rash and arterial hypertension. The types and frequency of adverse events observed in this study that led to discontinuation of leflunomide were very similar to those reported

Table VI. Reasons for discontinuation of leflunomide in patients taking different treatment strategies

Reason for discontinuation	Leflunomide monotherapy [n (%)] (n = 52)	Leflunomide and methotrexate [n (%)] (n = 17)	Leflunomide and other DMARDs ^a [n (%)] (n = 9)	Leflunomide and TNF α inhibitors [n (%)] (n = 12)
Total drug discontinuations	20 (38.5)	3 (17.6)	2 (22.2)	5 (41.7)
Lack of efficacy	7 (13.5)	1 (5.8)	0	2 (16.6)
Adverse effects	14 (26.9)	2 (11.7)	2 (22.2)	3 (25.0)
gastrointestinal ^b	6 (11.5)	2 (11.7)	0	2 (16.6)
arterial hypertension ^c	3 (5.7)	0	1 (11.1)	0
cutaneous ^d	5 (9.6)	1 (5.8)	1 (11.1)	1 (8.3)
others ^e	1 (1.9)	0	0	0

a DMARDs other than methotrexate (i.e. chloroquine, ciclosporin and salazopyrine).

b Diarrhoea, hepatotoxicity, nausea, gastric pain, vomiting.

c New onset.

d Skin rash, hair loss, angioedema.

e Polyneuropathy.

DMARDs = disease-modifying antirheumatic drugs; **TNF** = tumour necrosis factor.

from post-release clinical experience.^[3] In our study, we did not observe significant differences in the manifestation of adverse events between young and elderly RA patients. Hypertension is a common problem in elderly subjects aged ≥ 60 years, reaching a prevalence as high as 60–80%.^[15] A small percentage of patients with RA develop hypertension during leflunomide therapy, especially in the first months of treatment, and no other serious cardiovascular adverse effects have been reported in the literature.^[16] Few patients in our cohort discontinued treatment because of hypertension, and there was no significant difference in this respect between younger and older patient groups. Concurrent therapy with NSAIDs is a risk factor, and all patients in our study were taking an NSAID.

Three-fold elevation of serum aminotransferases has been noted in up to 13% of patients treated with leflunomide but is generally reversible with dose reduction or discontinuation of the drug.^[16] All patients in our study who experienced nausea and diarrhoea or elevation of aminotransaminases were taking concomitant NSAIDs or methotrexate therapy, which might have contributed to hepatotoxicity. These patients had no history of previous or concurrent alcohol abuse, or of viral or autoimmune hepatitis. Rash and hair loss have been noted in 10–15% of patients taking leflunomide,^[17] compared with a rate of 8.8% in our study. The results of our analysis confirmed a similar drug discontinuation rate in elderly compared with younger patients. Furthermore, we obtained the same finding when we compared drug discontinuation rates in patients receiving leflunomide monotherapy with those receiving combination therapy.

Our study confirms that a higher proportion of older patients with RA receive leflunomide as monotherapy, whereas leflunomide is more often combined with methotrexate, other DMARDs or TNF α inhibitors in younger patients. Non-compliance, risk of dosage errors and increased incidence of polypharmacy can reduce drug prescribing by rheumatologists to elderly patients. Compared with younger patients, older patients tend to have decreased cardiac output, decreased renal and hepatic blood flow, decreased total

body water and increased body fat, which can affect drug distribution in patients taking multiple medications and result in toxic drug concentrations.^[8] Our results confirmed that elderly RA patients receive combination therapeutic strategies less commonly, a finding that reflects physician concerns that patients in this group are more likely to have purified protein derivative positivity, malignancy in remission or underlying cardiac dysfunction. Our analysis showed that, in patients aged >65 years, leflunomide in combination with other DMARDs or TNF α inhibitors appeared to be well tolerated and maintained over time. Lack of efficacy was more frequent with leflunomide alone than when combined with another therapy.

Conclusion

Our analysis showed that the safety profile of, and adherence to, leflunomide therapy in older patients with chronic inflammatory joint diseases such as RA or psoriatic arthritis is no different to that observed in younger patients.

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